Goal Modeling

Radiation Therapy Controller

- We are going to immerse ourselves into something called reading.
- Work with a group around you and read the prompt on Canvas for the Radiation Therapy Controller.
- Take notes on the following
  - Agents / actors
  - At least 5 functional requirements you envision for the
- You have 10 minutes.

Goals, Software Requirement, Dom assumptions

- A goal is a statement
  - Formulated entirely in terms of World phenomena
  - To be enforced by Machine in collaboration with other agents in the World
  - Synonyms: system requirement, system goal, business goal
- A software requirement is a statement
  - Formulated entirely in terms of shared phenomena
  - To be enforced by the Machine alone
  - Synonyms: software specification
- A domain assumption is a statement
  - Formulated entirely in terms of World phenomena
  - To be enforced by World components other than the Machine

Jan 27 Poll

Join Code: S47XBL

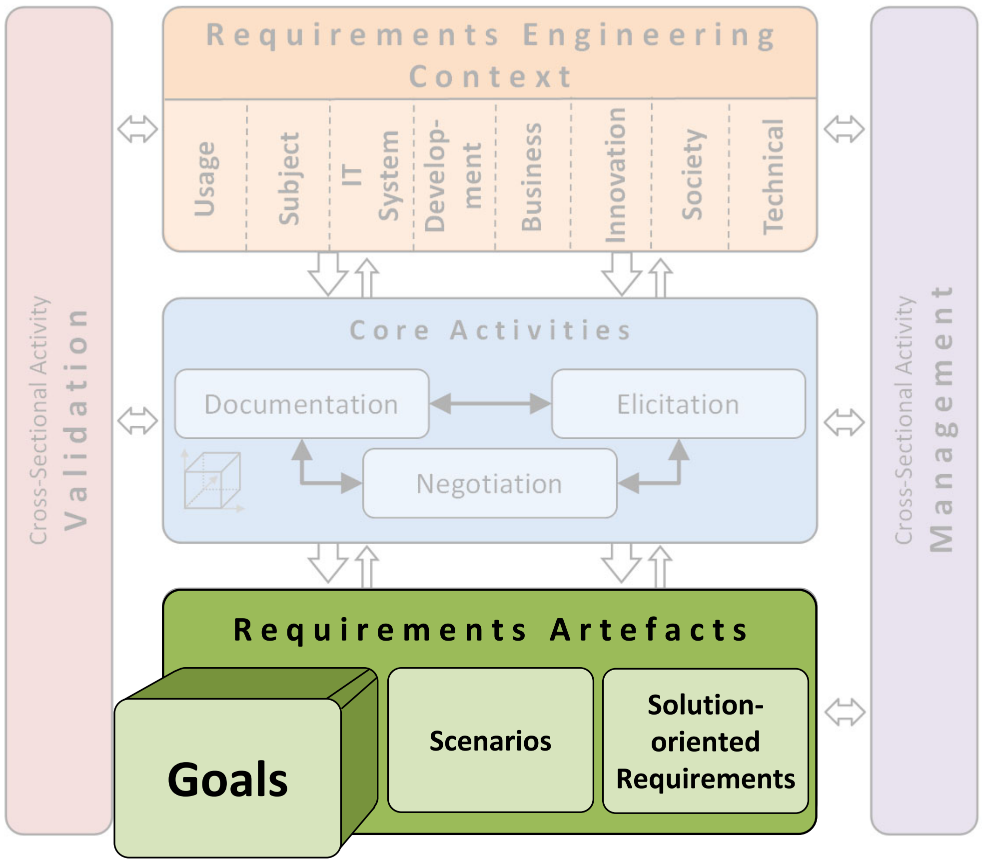
Nebraska App

1. Navigate to Polls
2. Choose this poll from the list or enter the join code

Web Browser

1. Go to go.unl.edu/polling
2. Choose this poll from the list or enter the join code

Framework for Requirements Engineering



Fundamentals

Definition

D A goal describes an **intention** of one or more **stakeholders** with regard to the **objectives**, **properties** or **use** of the **system**.

- **Characteristics of a goal:**
- **Refines the system vision** into more detailed objectives to be fulfilled by the system
- Should be **solution free**

- In **agile** development, **epics** are used as a kind of **high-level goals**

Fundamentals

Goals: Examples

E Car navigation system:

- G1: The system shall automatically guide the driver to a desired destination.
- G2: The response time of the system shall be 20% shorter compared to the predecessor system.
- ...

E University library system:

- G2 : The system shall provide functionalities for students and employees.
- G5: The system shall allow employees to use the library printers directly from their work stations.
- ...

Fundamentals

Benefits of Using Goals (1)

- **Foster system understanding**
  - Goals refine the system vision, clarify the value of the system, and provide rationales for developing the system.
- **Requirement elicitation**
  - Goals drive and guide the elicitation of requirements, e.g., a set of requirements can be defined which must be fulfilled to satisfy the goal, or scenarios can be used to document typical goal satisfaction interactions.
- **Identification and evaluation of alternative realisations**
  - Decomposing goals into sub-goals supports the identification of alternative solutions and their evaluation.
- **Justifcations of requirements**
  - If a requirement contributes to the satisfaction of a goal, the **goal provides a rationale** for defining a requirement.

Fundamentals

Benefits of Using Goals (2)

- **Detection of irrelevant requirements**
  - If a requirement does not contribute to the satisfaction of any defined goal, either the **requirement is irrelevant** for the system or the **goals defined are incomplete**.
- **Completeness of requirements specifications**
  - If by realising the specified requirements all defined goals (stakeholder intensions) can be satisfied, the specification is complete.
  - If a goal cannot be satisfied by realising the requirements, **requirements are missing**
- **Identification and resolution of conflicts**
  - Focus on **solving goal conflicts first** (which document different stakeholder intentions), instead of solving conflicts between more detailed requirements artefacts (which result from different stakeholder intentions)
- **Stability of goals**
  - Goals are typically **more stable** than solution-oriented requirements.

AND/OR Goal Decompositions

AND-Decomposition

D The decomposition of a super-goal G into a set of sub-goals $G_1, \ldots, G_n$ with $n \geq 2$ is an AND-decomposition if and only if **all sub-goals** $G_1, \ldots, G_n$ **must be satisfied** in order **to satisfy the super-goal** G.

E Goals defined for a navigation system:

G_1 : Comfortable and fast navigation to the destination

The goal G_1 is **decomposed** into three sub-goals by means of an **AND-decomposition**:

$G_{1.1}$: Automatic navigation according to user-specific parameters

$G_{1.2}$: Automatic re-routing to avoid traffic jams

$G_{1.3}$: Easy entry of the destination

OR-Decomposition

D The decomposition of a super-goal G into a set of sub-goals $G_1, \ldots, G_n$ with $n \geq 2$ is an OR-decomposition if and only if **satisfying one of the sub-goals** $G_1, \ldots, G_n$ **is sufficient for satisfying the super-goal** G.

E $G_{1.2}$: Automatic re-routing to avoid traffic jams

The goal $G_{1.2}$ is **decomposed** by an **OR-decomposition** into two sub-goals:

$G_{1.2.1}$: Manual entry of traffic jams in road traffic

$G_{1.2.2}$: Autonomous update of traffic data

AND/OR Goal Trees: Definition

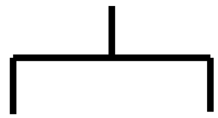
D An AND/OR goal **tree** consists of **nodes** that represent **goals**, and **directed edges** that represent **AND-decomposition** and **OR-decomposition** relationships between the goals. Each node (except of the root node) is related to **exactly one super-goal**.

Characteristics of a goal tree:

- Each goal can only **contribute to one super-goal**;
  a goal can thus have at most **one incoming edge**
- **No cyclic paths**; a goal cannot be directly or indirectly related to itself
- Goals are usually denoted using **short labels**;
  Textual annotations can be used to document details about a goal

AND/OR Goal Trees and Graphs

Graphical Notation for AND-Decomposition

Graphical notation: 

E

G_1: Comfortable and fast navigation to destination

$G_{1.1}$: Automatic navigation

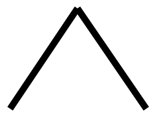
$G_{1.2}$: Circumnavigating traffic jams

$G_{1.3}$: Easy entry of destination

Note: The **direction** of an edge is **not explicitly shown** in the graphical notation; AND/OR goal trees are **read from top to bottom** (root goal at the top)

AND/OR Goal Trees and Graphs

Graphical Notation for OR-Decomposition

Graphical notation: 

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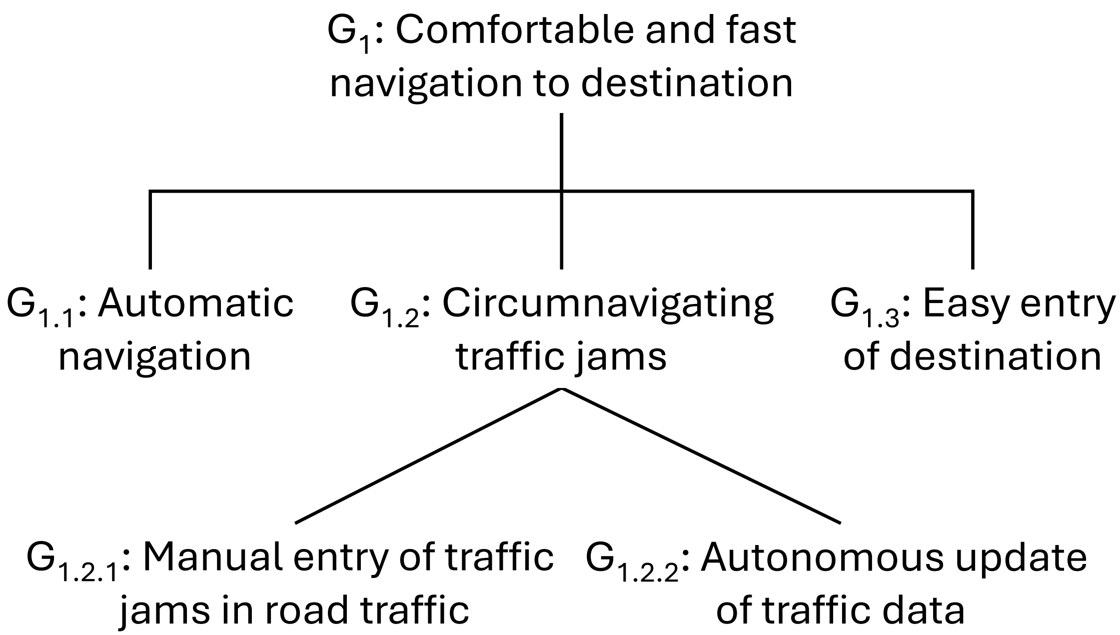
$G_{1.2}$: Circumnavigating traffic jams

$G_{1.2.1}$: Manual entry of traffic jams in road traffic

$G_{1.2.2}$: Autonomous update of traffic data

AND/OR Goal Trees and Graphs

AND/OR Goal Trees: Example



AND/OR Goal Graphs: Definition

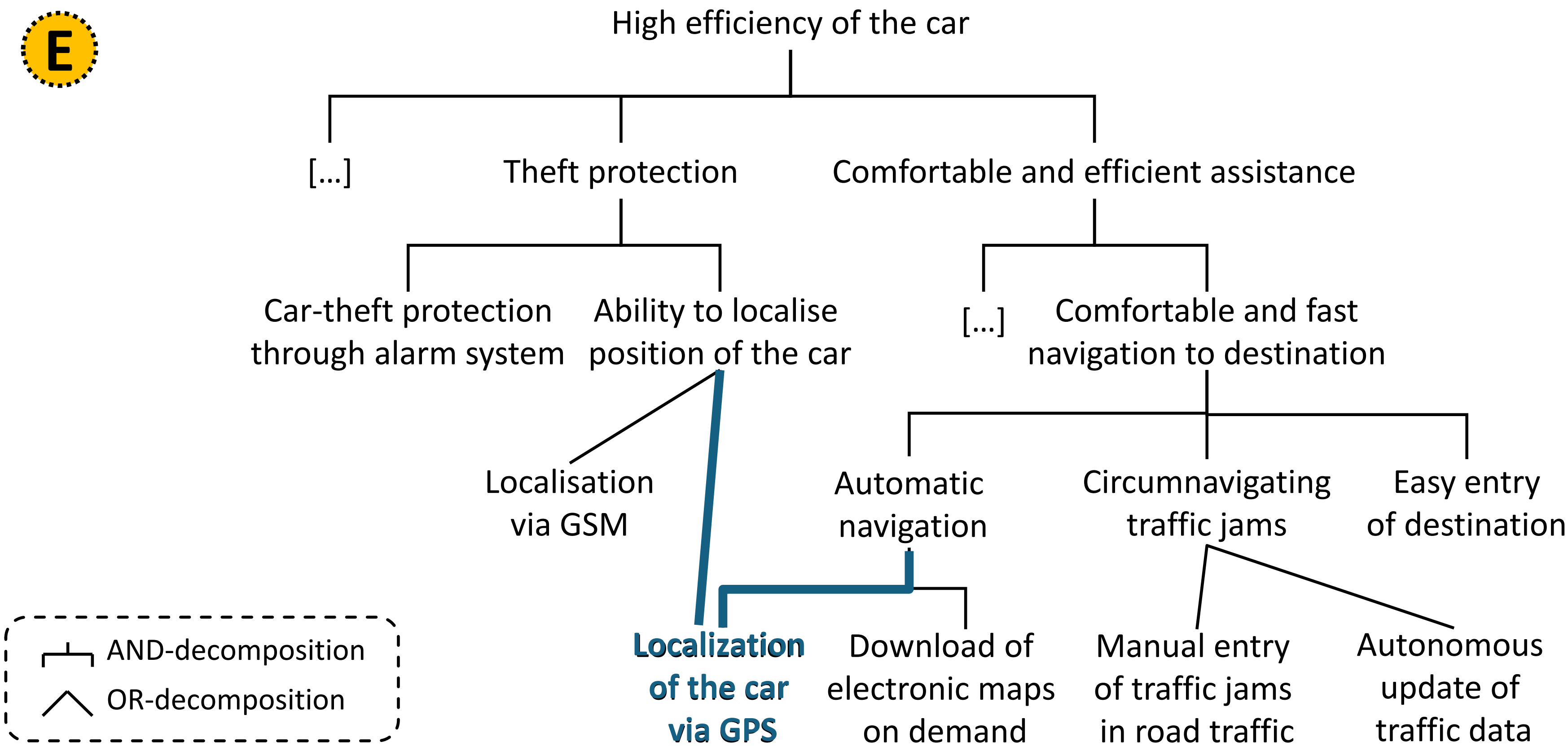
In practice, AND/OR goal trees are often **too restrictive**:
A **sub-goal often contributes to** the satisfaction of **more than one super-goal**

D An AND/OR goal graph is a **directed, acyclic graph** with **nodes representing goals** and **edges** representing **AND-decomposition** relationships and **OR-decomposition** relationships between the goals.

- A node in an AND/OR goal graph can have **more than one incoming edge** (in contrast to AND/OR trees)
- AND/OR graphs are **acyclic**

AND/OR Goal Trees and Graphs

AND/OR Goal Graphs: Example

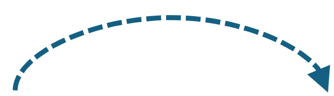


Five Common Goal Dependencies

Requires Dependency (1)

D A goal G_1 requires a goal G_2 if the **satisfaction** of G_2 **is a prerequisite** for satisfying G_1.

Common notation for requires dependencies:



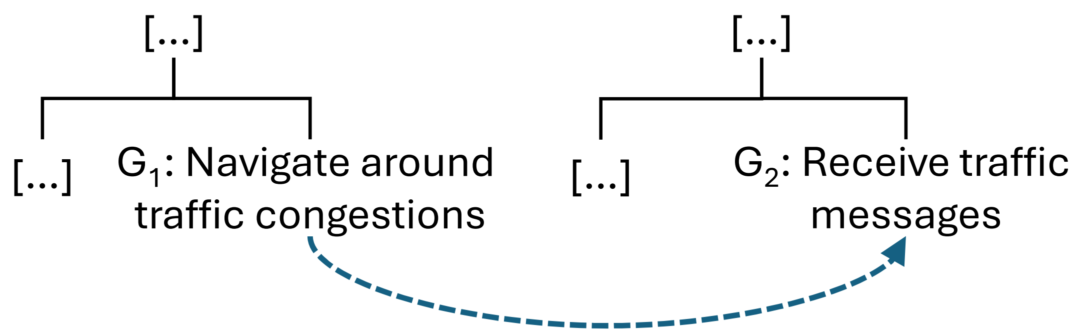
Five Common Goal Dependencies

Requires Dependency (2)

- G1: The system shall navigate the driver around traffic congestions.
- G2: The system shall be able to receive traffic messages.

The capability of receiving traffic massages is a prerequisite for calculating a route around traffic congestions

Requires dependency: G_1 requires G_2



Five Common Goal Dependencies

Conflict Dependency (1)

D A conflict exists between a goal G_1 and a goal G_2 if

(1) **Satisfying** G_1 **excludes the satisfaction** of G_2 **and**

(2) **Satisfying** G_2 **excludes the satisfaction** of G_1.

Common notation for conflict dependencies:

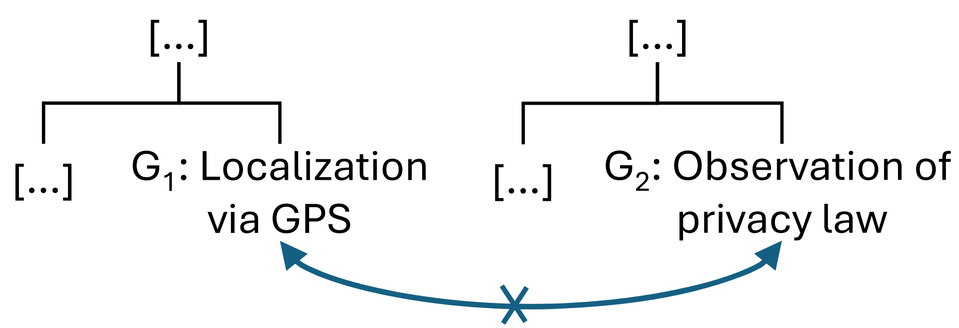
Five Common Goal Dependencies

Conflict Dependency (2)

- G1: It shall be possible to localize the car via GPS.
- G2: The country-specific privacy laws shall be observed.

The privacy law forbids the localization of cars via GSP by third parties

Conflict dependency: G_1 and G_2 are conflicting

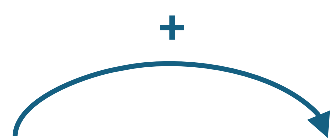


Five Common Goal Dependencies

Support Dependency (1)

D A goal G_1 supports a goal G_2 if the **satisfaction** of G_1 **contributes positively** to satisfying G_2.

Common notation for support dependencies:



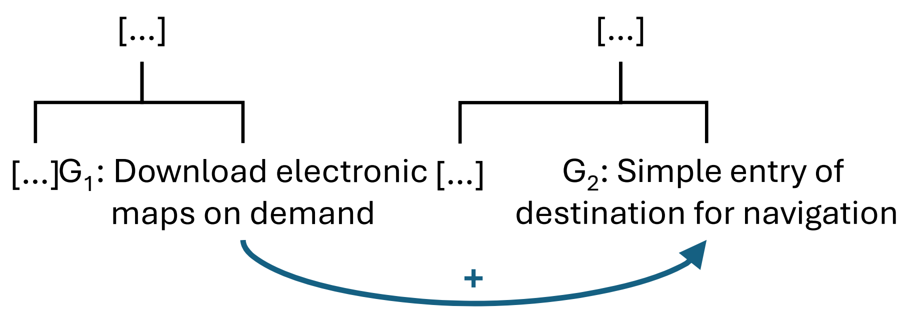
Five Common Goal Dependencies

Support Dependency (2)

- G_1 : The navigation system shall be able to download electronic maps on demand.
- G_2 : The entry of destination shall be simple.

Download of electronic maps helps the driver to enter a new destination in an area not covered by the previously available maps

Support dependency: G_1 supports G_2

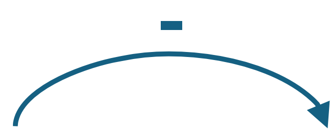


Five Common Goal Dependencies

Obstruction Dependency (1)

D A goal G_1 obstructs a goal G_2 if **satisfying** G_1 **hinders the satisfaction** of G_2.

Common notation for obstruction dependencies:



Five Common Goal Dependencies

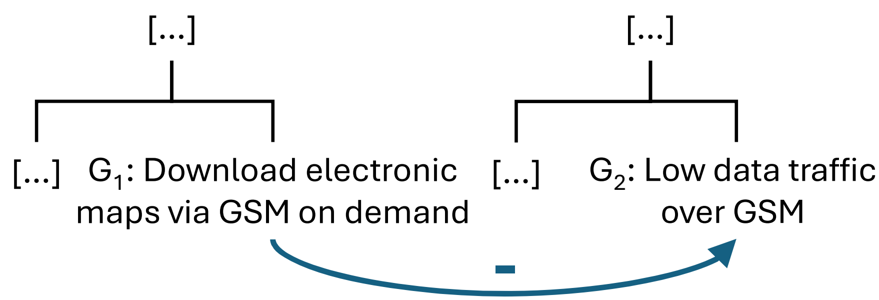
Obstruction Dependency (2)

E

- G1: The navigation system shall be able to download electronic maps via the GSM network on demand.
- G2: The data traffic over the GSM network caused by the navigation system shall be as low as possible.

Downloading electronic maps via GSM network causes considerable additional GSM data traffic

Obstruction dependency: G_1 interferes with G_2

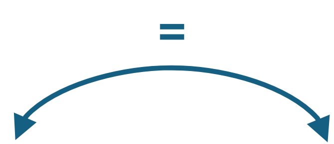


Five Common Goal Dependencies

Equivalence Dependency (1)

D Two goals G_1 and G_2 are equivalent (with respect to goal satisfaction) if **satisfying** the goal G_1 **leads to the satisfaction** of the Goal G_2 **and satisfying** the goal G_2 **leads to the satisfaction** of the goal G_1.

Common notation for equivalence dependencies:



Five Common Goal Dependencies

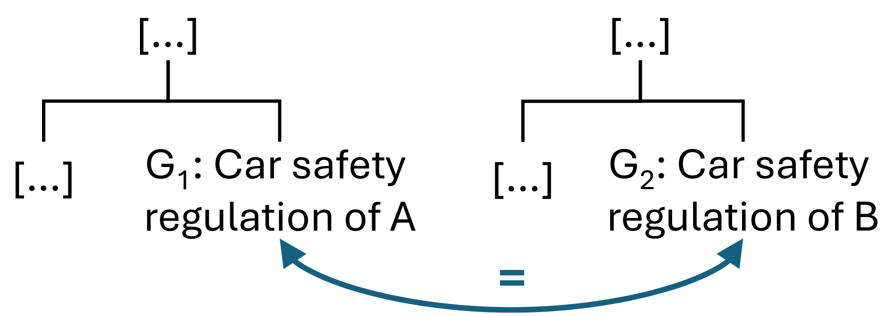
Equivalence Dependency (2)

E

- G1: The system shall comply with the car safety regulations of country A.
- G2: The system shall comply with the car safety regulations of country B.

The safety regulations of country A and country B are identical
(e.g., country A and country B define the safety regulations jointly for both countries)

Equivalence dependency: G_1 and G_2 are equivalent



Five Common Goal Dependencies

Context Impact on Goal Dependencies

- The **identification** and the **documentation** of goal dependencies strongly depends on the **context**,
  e.g., goal dependencies are context dependent (privacy law forbid localisation of cars in some countries)
- **Changes** in the **context** (e.g., change of a law) and/or changes in the **information known about the context** can affect the documented goals and their dependencies
- A change in the context requires a review of whether the documented **goal dependencies** and even the **goals themselves are still valid**

Textual Goal Documentation – Seven Rules

Rule 1: Document Goals Concisely

- Document goals **as concisely as possible** (but **not too briefly**)
- **Avoid** unnecessary phrases, filler and repetition

E

Goal G1:

Expert users as well as inexperienced users shall be able to use the system. Inexperienced users shall be able to use the system without having knowledge about the predecessor system. Furthermore, an inexperienced user shall be able to use the system without any training. For any user, it must be self-evident how to use the system. It must be possible to use the system even without knowledge of similar systems.

More concise textual definition of G1:

The users shall be able to use the system without training and/or knowledge of the previous system.

Rule 1

Rule 2: Use Active Voice

- **Avoid** using the passive voice; use **active voice** instead!
- Active voice **enhances understandability** and clearly names the **actor**

E

Goal G2:

The duration of creating the quarterly reports shall be cut down by half compared with the predecessor system.

Definition of G2 using active voice:

The user shall be able to create quarterly reports in half of the time required when using the current system.

Rule 2

Textual Goal Documentation – Seven Rules

Rule 3: Document the Intention Precisely

- Document the stakeholder's **intention precisely** (as far as possible)
- It should be possible to (later) **objectively check** whether the implemented system **satisfies the goal or not**
- However, objectively verifiable documentation of a goal is **not always desirable** or **possible**

E

Goal G3:

The system shall lead to an improved workflow in the company.

Definition of G3 with a precise intention:

The system shall speed up the workflow for order processing to decrease costs by at least 20% according to management.

Rule 3

Rule 4: Decompose High-level Goals

- An abstract goal should be **decomposed into sub-goals**
- The specific sub-goals can be **validated more easily**

E

Goal G4:
Increase driving safety.

G4 is decomposed into sub-goals by means of an AND-decomposition:

- G4.1: Reduce the braking distance on slippery roads by 20%.
- G4.2: Ensure that the vehicle remains steerable during braking maneuvres.

Rule 4

Rule 5: Document the Added Value

- Describe the **added value** associated with the goal as precisely as possible

E

Goal G5:

The navigation system shall provide an intuitive way of entering the destination of a trip.

Definition of G5 with explicit statement of expected added value:

The navigation system should offer a simple, unobtrusive way of entering the desired destination **so as not to distract the driver from driving**.

Rule 5

Rule 6: Document the Reasons for Introducing a Goal

- Provide a **brief and precise description** of the **reasons** for introducing the goal

E

Goal G6:

The system shall offer an intuitive user interface.

Definition of G6 including rationales:

The system shall offer an intuitive user interface to support the approx. 80% of the users, which use the system only once or twice a month.

Rule 6

Rule 7: Avoid Defining Unnecessary Restrictions (1)

- **Avoid** defining **unnecessary restrictions** which **constrain potential realisation** of the goal

E

Goal G7:

The response times of the system shall be reduced by 10% by optimizing the time for data transfers.

Definition of G7 without unnecessary restriction:

The response times of the system shall be reduced by 10%.

Rule 7

Rule 7: Avoid Defining Unnecessary Restrictions (2)

If a stakeholder (e.g., a client) demands a specific solution or expresses a specific constraint **restricting the realisation of the goal**, do the following:

- **Elicit** the **solution/constraint-free** (super) **goal** behind the restriction **by asking "why"** questions
- Try to **identify** viable **solution alternatives** for the constraint-free goal
- **Document** the identified, alternative solutions as sub-goals **using an OR-decomposition**
- **Discuss** the alternative sub goals with the stakeholder to validate if
  - The solution alternatives are viable as well
  - The originally proposed solution is really the desired one

Unstructured vs. Template-Based

- Goals and goal dependencies can be documented textually using:
- **Unstructured natural language**
- **Goal templates** that define **slots (attributes)** to document information about a goal (type), i.e., goal templates support the structured textual documentation of goals

- **Template-based documentation** of goals offers **significant advantages** compared to unstructured text-based goal documentation

Template-Based Goal Documentation

Characteristics of Goal Templates

- A **goal template** facilitates the **structured documentation** of information related to a goal
- A template defines **slots (attribute types)** and predefined values for the slots
- Templates are typically organised in a **tabular form**
- A goal templates could **differ** (e.g., mandatory/optional slots, slots defined), e.g., depending on
  - The project phases (e.g., early elicitation, requirements specification)
  - The project- and product specific needs
  - …

More about templates and attribute types/values when we talk about documentation in later weeks.

Template-Based Goal Documentation

Example of a Comprehensive Goal Template (1)

No.		Section	Content / Explanation
ID	1.1	Identifier	Unique identifier of the goal
	1.2	Name	Unique name for the goal
Management	2.1	Authors	Names of the authors who have documented the goal
	2.2	Version	Current version number of the documentation of the goal
	2.3	Change history	List of the changes applied to the documentation of the goal including (for each change) the date of the change, the version number, the author, and, if necessary, the reason for and the subject of the change
	2.4	Priority	Importance of the documented goal according to the specific prioritisation technique used
	2.5	Criticality	Criticality of the goal, for example, for the overall success of the system
Context	3.1	Source	Name of the source (i.e., the stakeholder, document, system, or data) from which the goal originates
	3.2	Responsible stakeholder	Name of the stakeholder who is responsible for the goal
	3.3	Stakeholders benefiting	Stakeholders who benefit from the satisfaction of the goal

Template-Based Goal Documentation

Example of a Comprehensive Goal Template (2)

No.		Section	Content / Explanation
Goal Definition	4.1	Goal level	Identifier for the abstraction level at which the goal is defined
	4.2	Goal description	Description of the goal
	4.3	Super-goal	Reference to the super-goal including the type of decomposition (AND/OR)
	4.4	Sub-goals	References to sub-goals including the type of decomposition (AND/OR)
	4.5	Other goal dependencies	Further dependencies with other goals such as “requires”, “conflict”, etc.
Relationships to other Artefacts	5.1	Use cases	Relationships of the goal to use cases
	5.2	Scenarios	Relationships of the goal to scenarios that describe the satisfaction of the goal or a failure to satisfy the goal
	5.3	Solution-oriented requirements	Relationships of the goal to solution-oriented requirements
	5.4	Other artefacts	Relationships of the goal to other artefacts, e.g., context information or design models
	6.	Remarks	Additional information about the goal (e.g., notes, open issues, etc.)

Template-Based Goal Documentation

Documentation of the Goal "Automatic Navigation" (Excerpt)

No.		Section	Content
ID	1.1	Identifier	G-2-17
	1.2	Name	Automatic navigation
Management	2.1	Authors	Peter Miller, Dan Smith
Context	3.1	Source	William Garland (product manager)
Goal Definition	4.2	Goal description	The system shall automatically direct the driver to the desired destination.
	4.3	Super-goal	G-2-2: Convenient and fast navigation to the destination (AND)
	4.4	Sub-goals	G-2-25: Localisation of the car via GPS (OR) G-2-26: Download of electronic maps on demand (OR)
	4.5	Other goal dependencies	Conflict with G-1-45: Reduce costs for cars Support of G-1-37: Technological leadership in the automotive segment of medium-sized vehicles

Template-Based Goal Documentation

Hints for Using Goal Templates

- Try to elicit **as many as possible relevant goals** first
- **Avoid** capturing **all attributes for a goal** right at the beginning; define **key attributes** first, e.g., ID, name, source, brief goal description
- **Subsequently**, identify **sub-goals** for each goal
- **Validate** whether the elicited goals are **complete** and the goal relationships are correct (Validation)
- **Complement** missing goals and missing goal relationships and, if required, **revise** the **goals** and **goal relationships** (LLMs are helpful)
- **Define scenarios** to support the elicitation and validation of goals (Journey maps and sequence diagrams)
- **Add missing information** in the goal template for all goals (LLMs are helpful)

Note: Use the slot "**Remarks**" for documenting information which does not fit in any other slot defined for the template, or **define additional slots**

Definition: Goal Model

D A goal model is a **conceptual model** that documents **goals**, their **decomposition** into sub-goals, and goal **dependencies**.

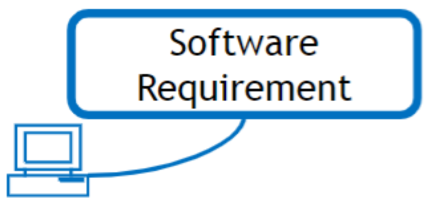
- A **goal modelling language** defines the modelling constructs (**syntax** and **semantics**) to be used **for creating a goal model**
- Goal models provide an **overview** of the **goals** and the goal **dependencies**
- Goal models support the **understanding** and **communication** about the goals
- In contrast to textual goal documentation, a **short label** is used to document a goal in a goal model
- For documenting goals, we recommend using both, **goal models** and **templates**

Another representation of goals

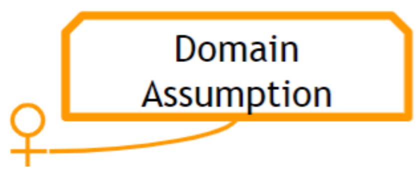
Something a stakeholder wants to be true of the world

Goals can be refined into subgoals

Goal refinement stops when leaf goals can be assigned to single agents, either software or environment



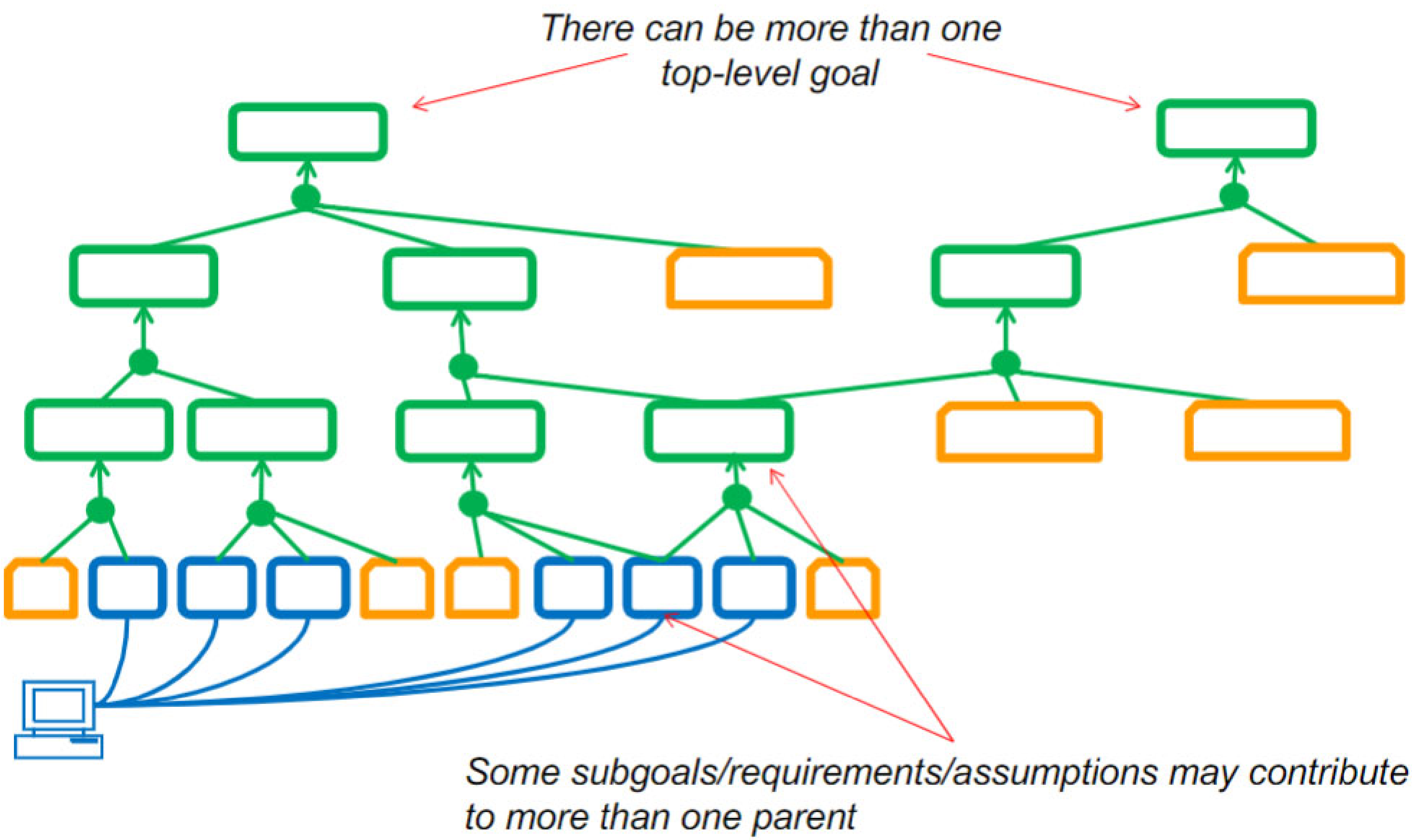
A goal under responsibility of the software



A goal under responsibility of environment agents

A typical goal model

- Can be written and read top-down and bottom-up (usually a combination)



Goal Example

Goal: Achieve [Ambulance Intervention]

Definition:

GIVEN an incident has occurred

WHEN the ambulance service receives a call reporting the incident

THEN within 14 minutes a first ambulance arrives at the incident scene

Defining an ACHIEVE goal

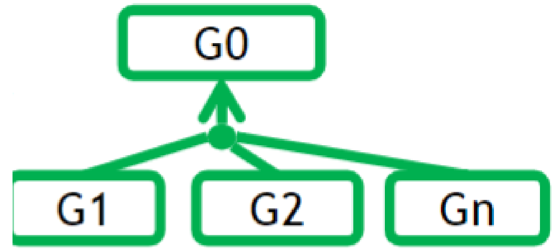
An Achieve goal prescribes intended behaviors where a target condition must sooner or later hold whenever some other condition holds in the current system state.

When we want to highlight the goal type, we will prefix the goal name with the corresponding keyword. For an Achieve goal we will write:

Achieve [TargetCondition]

And Refinement Links

- Semantic: parent goal can be satisfied by satisfying **all** subgoals



Important!

And-refinements should be

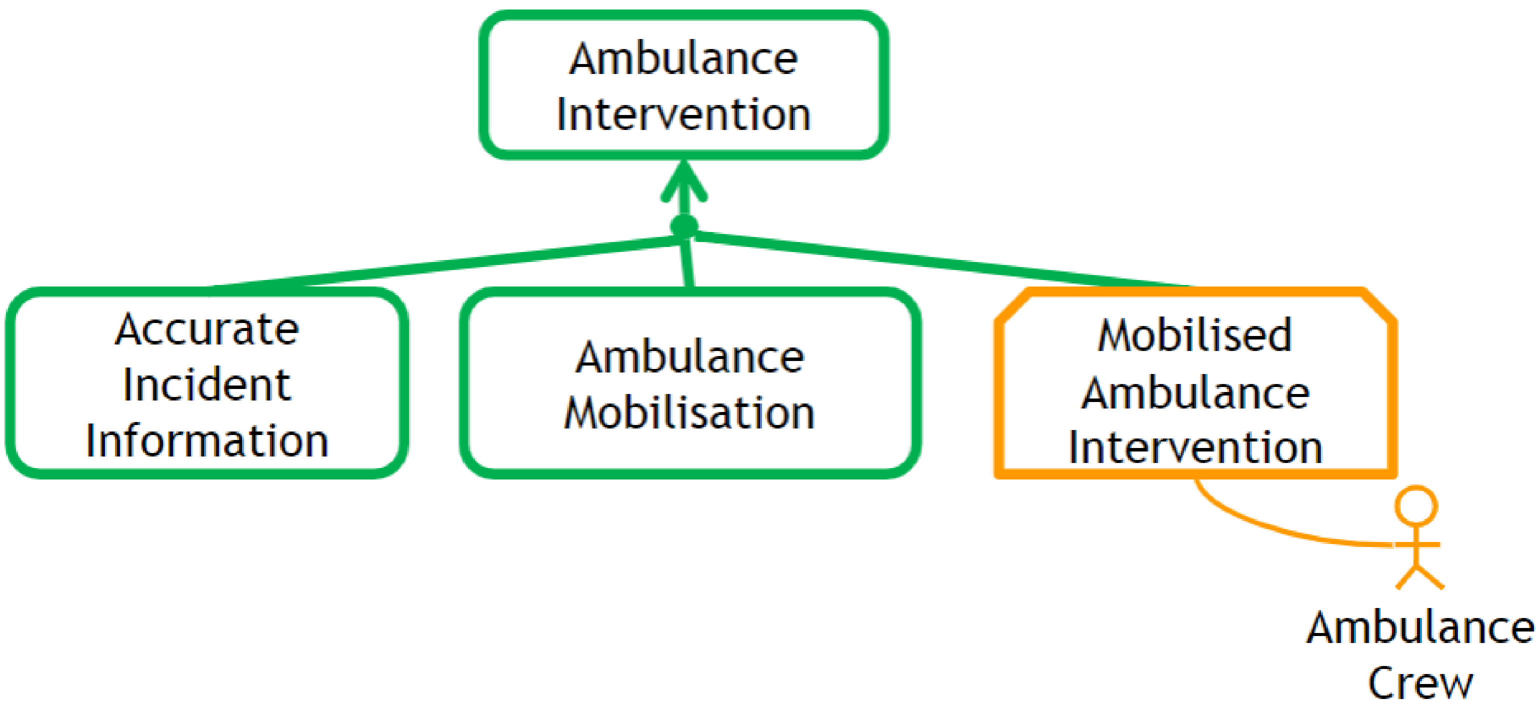
- Complete: $G1 \wedge G2 \wedge \ldots \wedge Gn \Rightarrow G0$
- Consistent: $G1 \wedge G2 \wedge \ldots \wedge Gn \not\Rightarrow \text{false}$
- Minimal: $\wedge_{\{j \text{ st. } j \neq i\}} Gj \not\Rightarrow G0$

(i.e. all subgoals are necessary)

Example of goal refinement

Think “problem decomposition”

Different subgoals concern different agents and world phenomena



Parent Goal

Goal: Achieve [Ambulance Intervention]

Definition:

GIVEN an incident has occurred

WHEN the ambulance service receives a call reporting the incident

THEN within 14 minutes a first ambulance arrives at the incident scene

Subgoals

Goal: Achieve [Accurate Incident Information]

Definition:

WHEN the ambulance service receives a call reporting the incident

THEN within 1 minute a call handler submits an incident form that has the accurate incident's location, number of causalities and their types.

Goal: Achieve [Ambulance Mobilization]

Definition:

WHEN a call handler submits an incident form

THEN within 2 minutes the available ambulance that is nearest to the location recorded on the incident form should be mobilized to that location

Domain Assumption

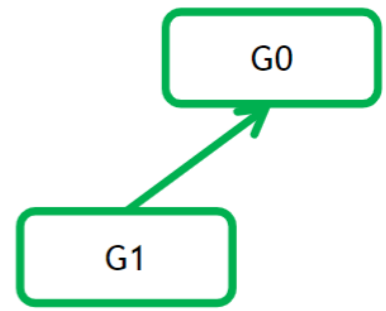
Assumption Achive[Mobilized Ambulance Intervention]

Responsibility Ambulance Crew

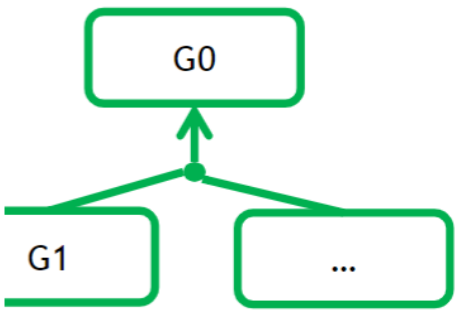
Definition

Given	ambulance 'a' is in service and available
And	ambulance 'a' is at location 'x'
When	ambulance 'a' is mobilised to go to location 'y'
Then	ambulance 'a' arrives at the location 'y' within the expected time needed to travel from 'x' to 'y'

Contribution Links



Semantics: a subgoal **contributes to** a parent goal if it is part of a (possibly unspecified) and-refinement of the parent goal.

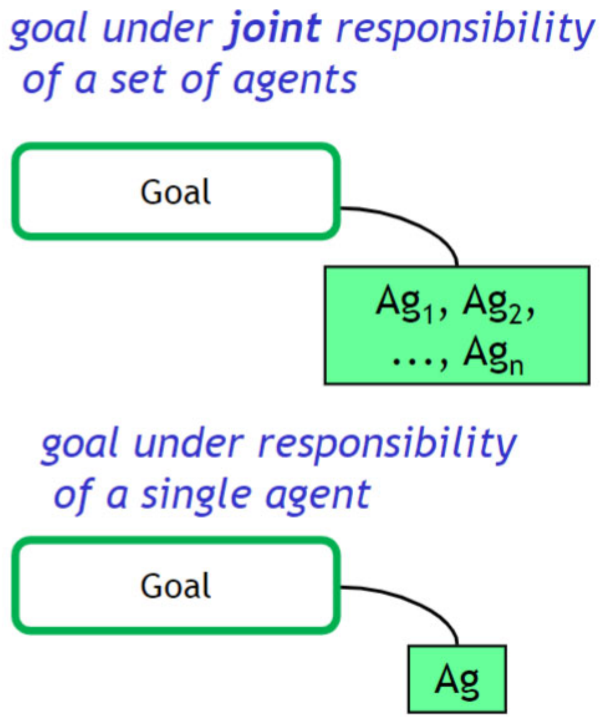


When to stop refining goals?

- Goals require the cooperation of agents for their satisfaction
  - Top level goals require the cooperation of many agents
  - Lower level goals require the cooperation of fewer agents
- Goals are refined into subgoals so that **each subgoal requires the cooperation of fewer agents** for its satisfaction
- Goal refinement **stops when leaf goal involves a single agent for its satisfaction**
  - Some leaf goals involve the software-to-be
  - Other leaf goals involve agents in the environment

Responsibility Links

- Semantics: an agent is **responsible** for a goal if it is required to restrict its behavior to satisfy the goal



Agents (Responsibility Links) and Context

- Every agent responsible for some goal must be included in the context diagram
- Every agent in the context diagram must be responsible for some goal
- Every requirement must be defined entirely in terms of machine phenomena

Example of a software requirement

REQ Achieve[Confirmed Allocation Sent to Ambulance MDT]

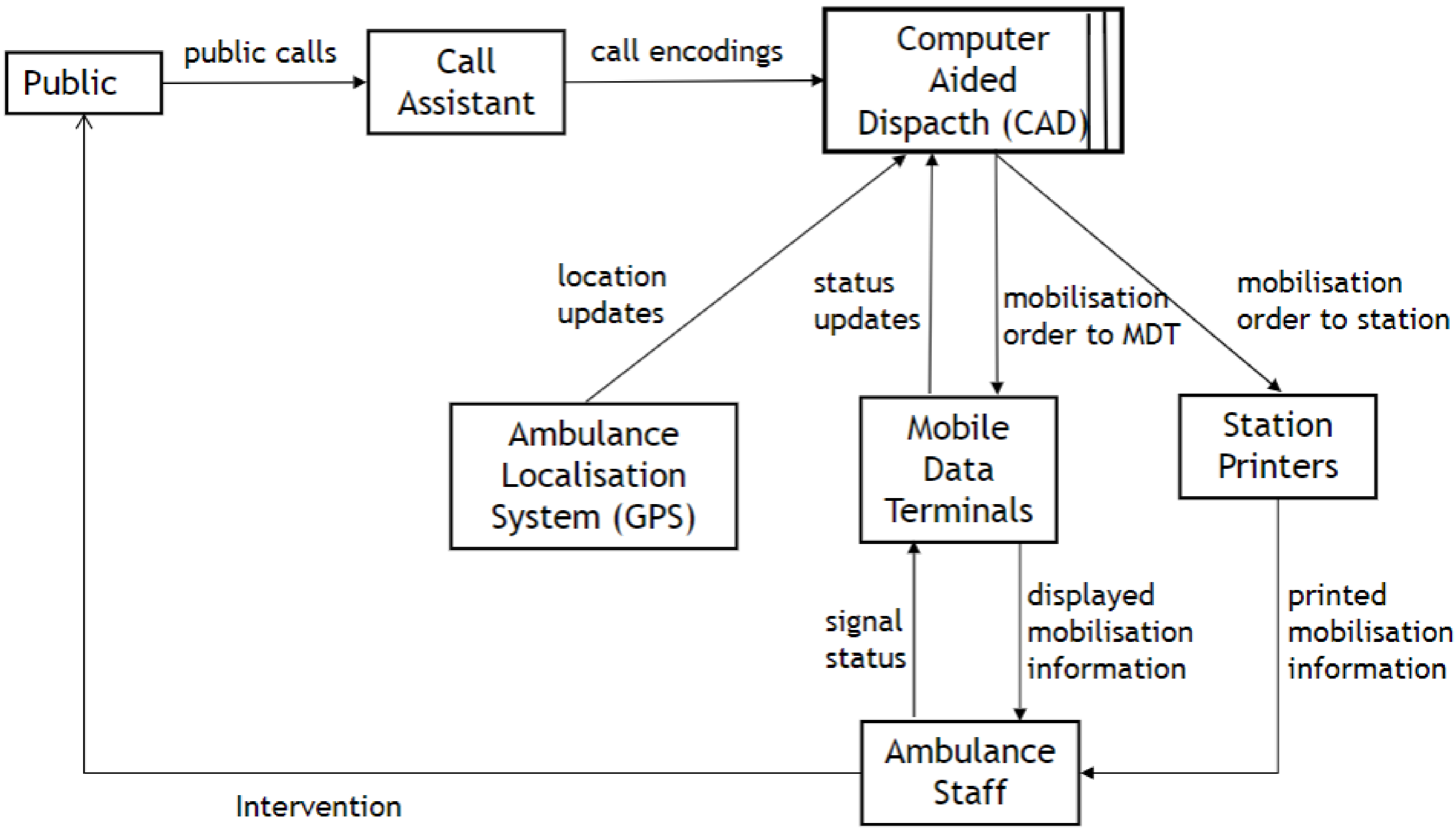
Responsibility Computer Aided Dispatch (CAD) Software

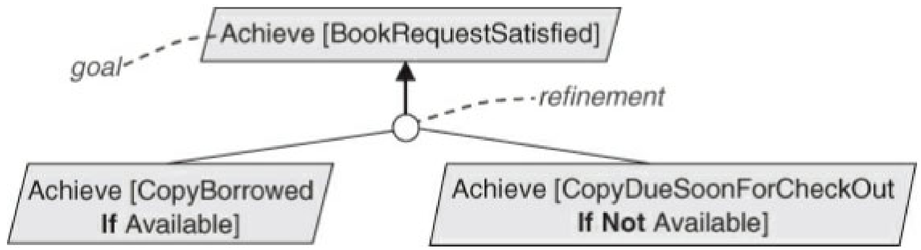
Definition

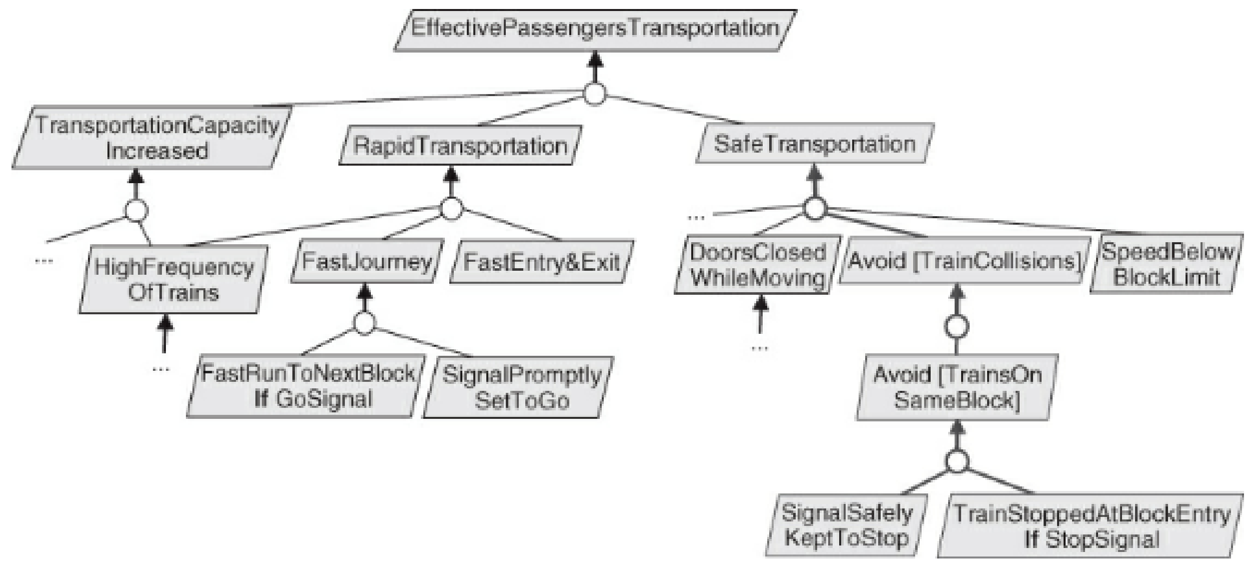
WHEN the Call Assistant confirms the allocation of an ambulance to an incident

THEN within 1 second the CAD shall send the mobilisation instructions to the ambulance’s Mobile Data Terminal (MDT)

A sneak peek into a Context Diagram of the CAD system




Achieve [BookRequestSatisfied]
goal
refinement
Achieve [CopyBorrowed If Available]
Achieve [CopyDueSoonForCheckOut If Not Available]


EffectivePassengersTransportation
TransportationCapacity Increased
RapidTransportation
SafeTransportation
HighFrequency OfTrains
FastJourney
FastEntry&Exit
DoorsClosed WhileMoving
Avoid [TrainCollisions]
SpeedBelow BlockLimit
FastRunToNextBlock If GoSignal
SignalPromptly SetToGo
Avoid [TrainsOn SameBlock]
SignalSafely KeptToStop
TrainStoppedAtBlockEntry If StopSignal

Refine this goal

Goal: Achieve [Ambulance Mobilization]

Definition:

WHEN a call handler submits an incident form

THEN within 2 minutes the available ambulance that is nearest to the location recorded on the incident form should be mobilized to that location

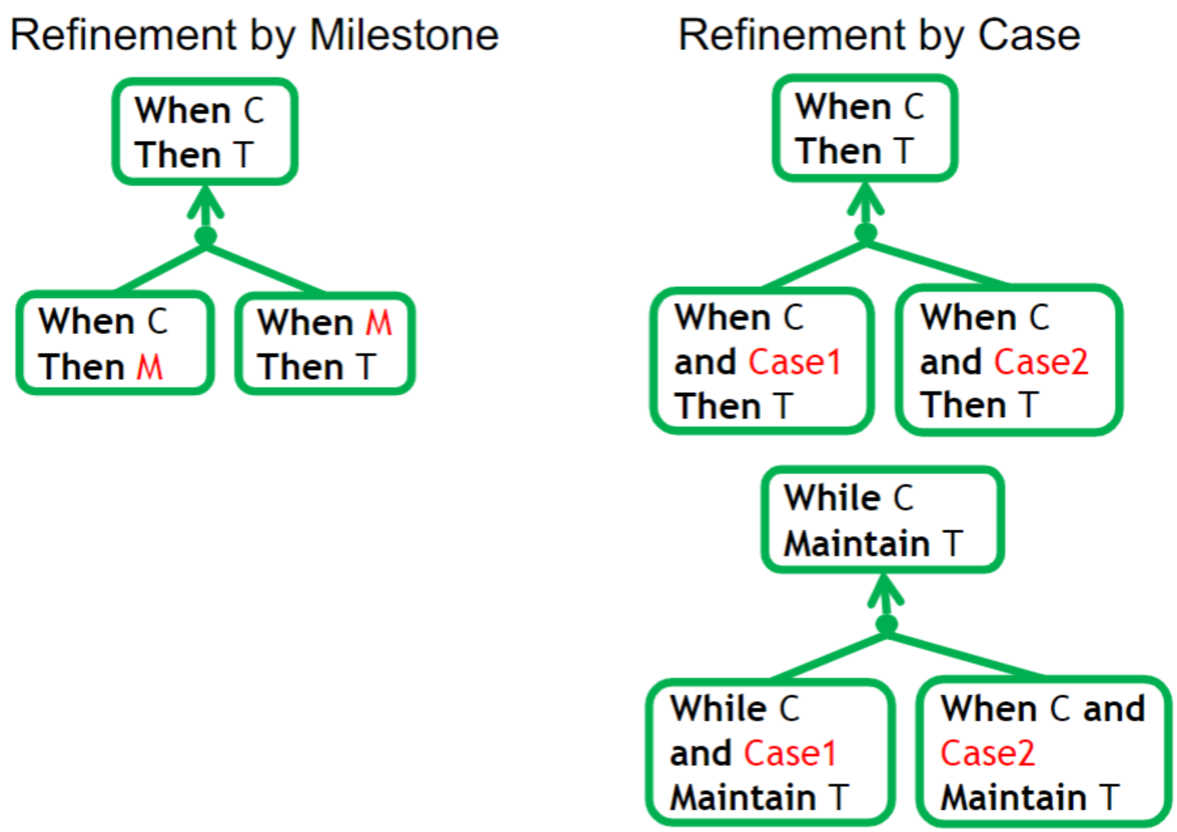
Practical Hints to Refine Goals

Refine goals using simple agent-driven strategy

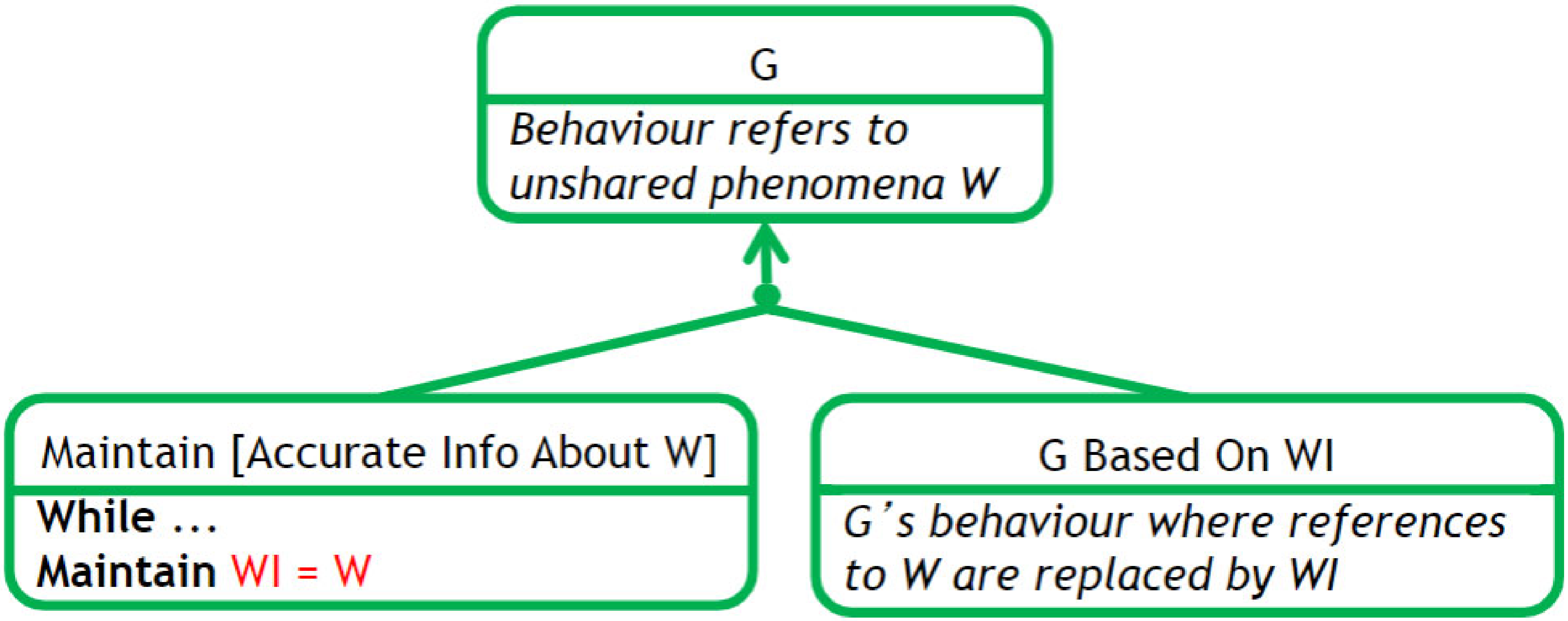
1. List all agents contributing towards satisfying the goal
2. For each agent in application domain, write assumptions needed to satisfy the goal
3. Write machine requirements that satisfy the goal with the above assumption

Refine goals using goal refinement patterns

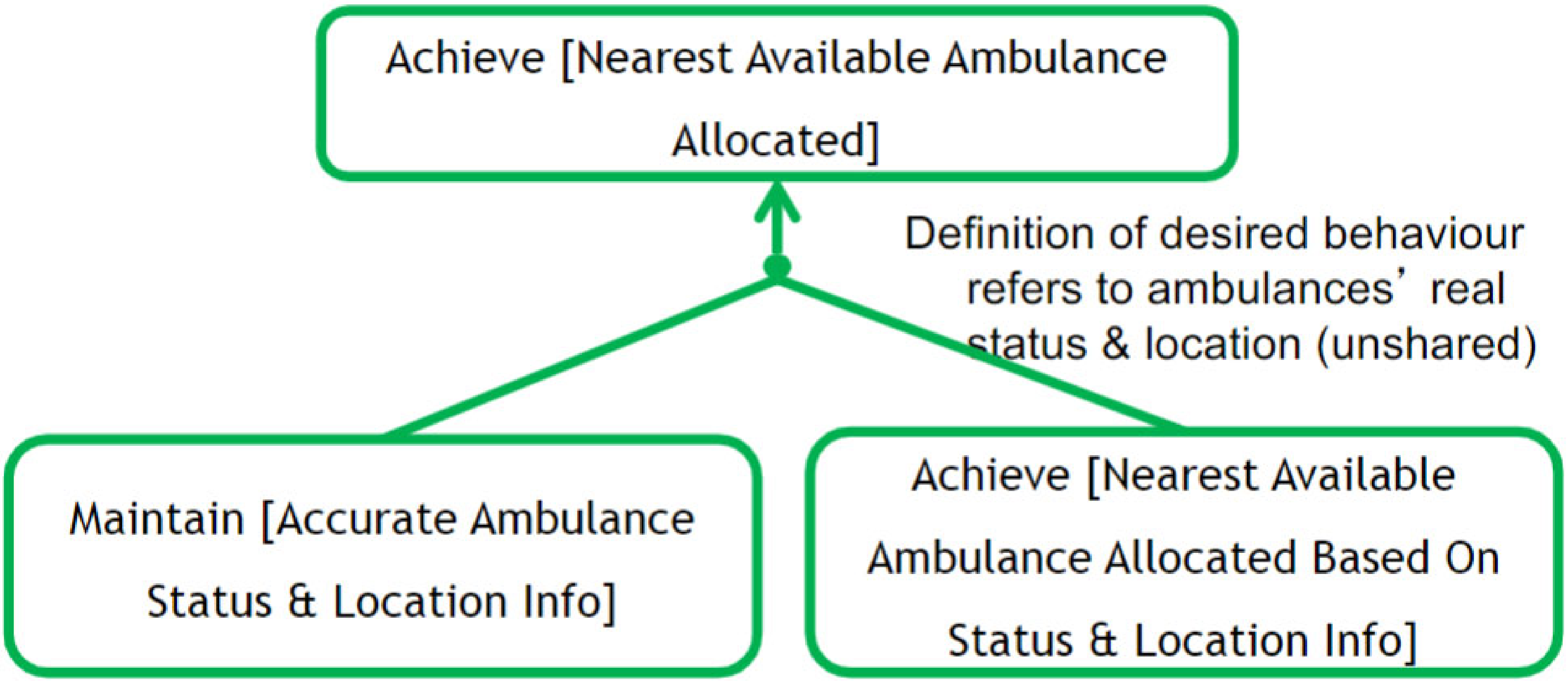
Two Useful Refinement Patterns



Third Useful Pattern: Introduce Accuracy Goal



Introduce Accuracy Goal - Example



Where do we even come up with these goals?

From documents, email exchanges, interviews, watching workflows, surveys, etc..

One excerpt from documents

The Personal Spine Information Services shall support the establishment and termination of care relationships between the patient and healthcare teams within organizations, and between the patient and individual Clinicians.

- Taken from the specification for the NHS Integrated Care Records Service (full spec has over 500 pages).

Sentence Patterns

A sentence pattern is a structure that can be used to express a variety of concrete goals, requirements and assumptions

Pattern

GIVEN <starting condition>
WHEN <trigger event>
THEN <resulting state>

Concrete Statement

GIVEN an incident has occurred
WHEN the ambulance service receives a call reporting the incident
THEN within 14 minutes
a first ambulance arrives at the incident scene

KAOS Sentence Patterns

Based on Given-When-Then
in Behaviour-Driven Development

Name	Pattern
Achieve	GIVEN <starting state> WHEN <triggering event> THEN <temporal qualification> <resulting state>
Maintain	WHILE <some state> MAINTAIN <Good condition>
Avoid	WHILE <some state> AVOID <Bad condition>

KAOS Statement Definitions

Statement type,
could be GOAL, REQ or DOM

pattern

Statement name

Goal Achieve [Ambulance Intervention]

Definition

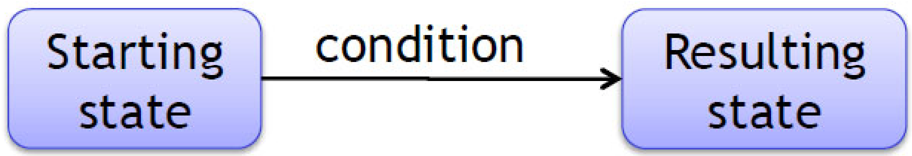
GIVEN an incident has occurred

WHEN the ambulance service receives a call reporting the incident

THEN within 14 minutes

a first ambulance arrives at the incident scene

Achieve Statements & Temporal qualification



If the system is in the starting state and the condition occurs, then the system will transition to the resulting state

Name	Pattern	Meaning
Immediate Transition	GIVEN <starting state> WHEN <trigger event> THEN <resulting state>	The transition is immediate
Achieve Eventually	GIVEN <starting state> WHEN <trigger event> THEN eventually <resulting state>	The transition will happen sometimes in the future (but there is no bound)
Bounded Achieve	GIVEN <starting state> WHEN <trigger event> THEN within <delay> <resulting state>	The transition will happen with the specified delay

Elevator System

Solution depends on context

- What is the machine? (e.g., control software)
- What are the shared phenomena? (the monitored and controlled phenomena)
- What are non-shared world phenomena?

Elevator examples (1)

Goal Achieve [Person at Destination Floor]

Definition

Given a person is at floor f

When the person wants to go to floor f’

Then eventually the person is at floor f’

Elevator examples (2)

REQ Achieve [Elevator request recorded]
Definition
Given the Elevator is not stopped at floor f
When a person at floor f presses the elevator request button
Then the elevator has a pending request for floor f

REQ Achieve [Elevator starts when doors closed]
Definition
Given the elevator is stopped at floor f
  And the elevator has a pending floor request
When the doors at floor f become closed
Then the elevator starts moving in its direction of travel

Maintain/Avoid Statements

Ambulance example

Goal Maintain [Accurate Ambulance Information]

Definition

While an ambulance is in service

Maintain the ambulance despatching software has accurate information about the ambulance status and location

Elevator example

Goal Avoid[Person Falling into Shaft]

Definition

Avoid someone falling into the elevator shaft

Related Patterns: EARS (created and used at Rolls Royce)

Name	Pattern	Example
Ubiquitous	The <system name> shall <system response>	The control system shall indicate the engine oil quantity to the aircraft
Event-driven	WHEN <trigger> <optional precondition>, the <system name> shall <system response>	When continuous ignition is commanded by the aircraft, the control system shall switch on continuous ignition
Unwanted Behaviour	IF <optional preconditions> <trigger>, THEN the <system name> shall <system response>	If the computed airspeed fault flag is set, then the control system shall used modelled airspeed
State-Driven	WHILE <system state>, the <system name> shall <system response>	While the aircraft is in-flight, the control system shall maintain fuel flow above x lbs/sec
Optional Feature	WHERE <feature is included>, the <system name> shall <system response>	Where a control component acts as a firewall, the component shall be Fireproof

A. Mavin, P. Wilkinson, BIG EARS The Return of the Easy Approach to Requirements Syntax, RE 2010

Related Patterns: Gherkin Scenarios

describe behaviour *examples* to be used as acceptance tests (more details in a later lecture)

Feature: Refund item

Scenario: Jeff returns a faulty microwave

Given Jeff has bought a microwave for $100

And he has a receipt

When he returns the microwave

Then Jeff should be refunded $100

Example from https://cucumber.io/docs/reference

Related Patterns: Formal Patterns of Temporal Logic Assertions (KAOS)

Name	Temporal assertion pattern
Achieve	$C \Rightarrow \Diamond T$
Bounded Achieve	$C \Rightarrow \Diamond_{\leq d} T$
Immediate Achieve	$C \Rightarrow \circ T$
Maintain	$\Box$ GoodState
Avoid	$\Box \neg$ BadState

- $\Box P$
  - P always hold
    - $\Box P$ holds at S_j iff P holds at all states S_k, $k \geq j$
- $\Diamond P$
  - P holds sometimes
    - $\Diamond P$ holds at S_j iff P holds at some state S_k, $k \geq j$
- $\circ P$
  - P holds at the next instant
    - $\circ P$ holds at S_j iff P holds at state S_{j+1}

Limitations of RE Theory

- Real requirements documents are never perfect. This theory does not help us assessing when a requirements document is “good enough”
  - an important unsolved research problem
- The theory describes the end result of a requirements elaboration process that involves exploring alternative domain assumptions and requirements. It does not support modeling and reasoning about such alternatives.
  - Extension to modeling and evaluating alternatives in goal models
- Satisfaction arguments deal principally with behavior requirements, not with quality requirements. The first are either satisfied or not satisfied; the second involves levels of satisfaction.
  - Extension to quantitative (probabilistic) goal models.

Summary

- A goal defines an intention of one or more stakeholders with regard to the objectives, properties or use of the system
- AND/OR Decomposition of goals
- Five common goal dependency types
- Seven rules for textual goal documentation
- Documenting goals with templates
- Conceptual goal models:
  - AND/OR trees and graphs: Simple, easy to understand visualisation of goals, goal decomposition and goal dependencies)
- Key benefits of goal-orientation

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